

Securing Harvest: Automated Detection and Classification of Rice Diseases Using DCNN



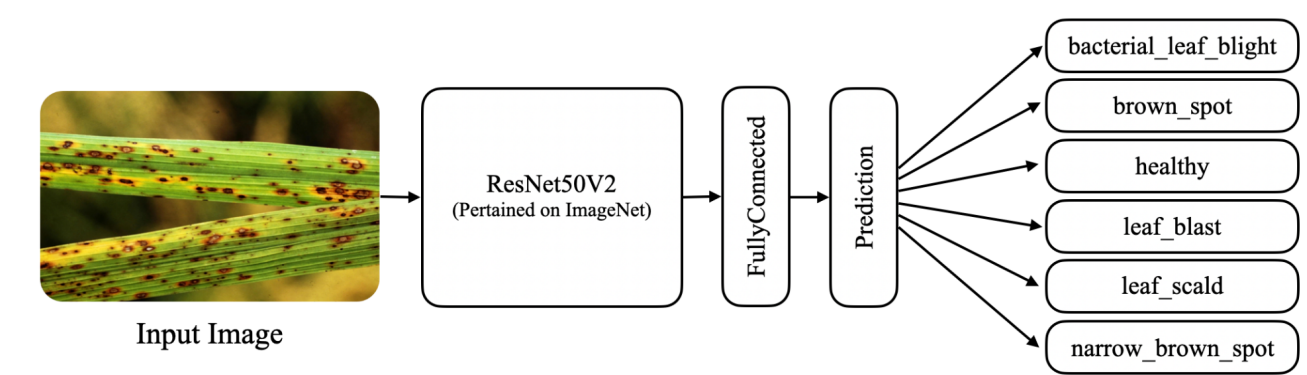
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Problem & Motivation

- **Problem Context:** Delayed, manual disease identification leads to crop loss, chemical misuse, and high dependency on scarce expert knowledge in regions like South Sudan.
- **Research Gap:** Existing deep learning models lack the accuracy, scalability, and resource efficiency required to run on low-power devices in diverse field conditions.
- **Objective:** Develop an automated, deep-learning system for accurate, early detection of rice leaf diseases directly from images to provide actionable farmer support.

Methodology & Dataset

System Architecture
Our system utilizes ResNet50V2-based Deep Convolutional Neural Networks.



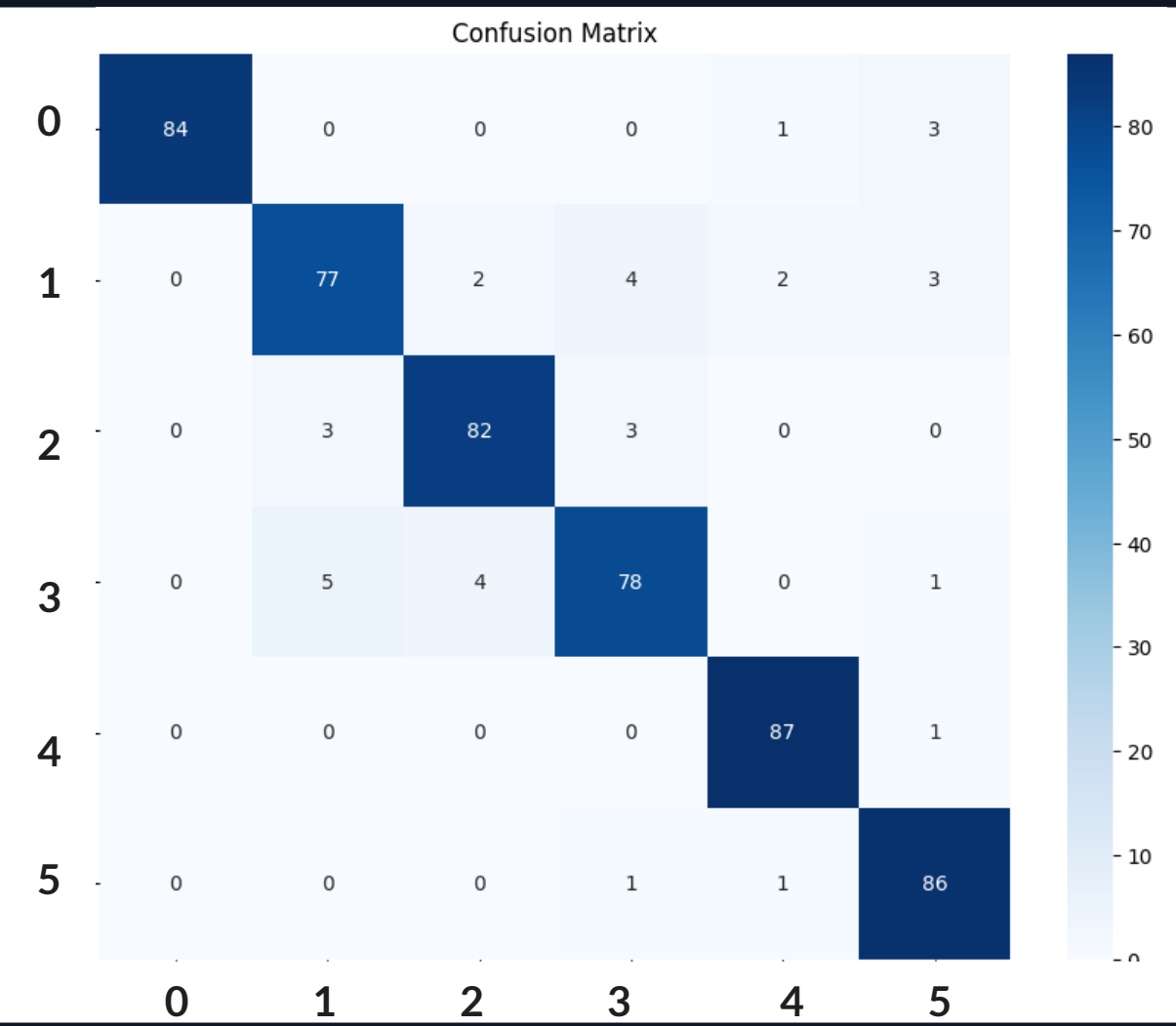
Dataset
Our data is rice leaf disease dataset, sourced from Kaggle consisting of: (3156+ images, 6 disease categories/classes).



Evaluation & Key Results

- Accuracy: 94.0%
- F1-Score (weighted): 94.0%

Class Index	Disease Name
0	Bacterial Leaf Blight
1	Brown Spot
2	Healthy
3	Leaf Blast
4	Leaf Scald
5	Narrow Brown Spot



Significance & Conclusion

- **Contribution:** Built a reliable, automated model that replaces slow, subjective manual disease identification.
- **Impact:** Deployed a mobile agricultural tool that provides instant field diagnosis and improves continuously via user feedback.
- **Future Work:** Conduct field testing, expand disease dataset categories, and roll out the full mobile application.